Course Syllabus for CSE-326

Part A

1. **Course Code:** CSE-326

2. **Course Title:** Internet Programming (Sessional)

3. **Course Type:** Core Course

4. **Level/Term:** Level-3, Term-I

5. **Academic Session:** 2020-21

6. **Course Teacher:** Ashim Dey, Assistant Professor, Dept. of CSE, CUET
   Md. Atiqul Islam Rizvi, Lecturer, Dept. of CSE, CUET

7. **Prerequisite(s):** None

8. **Credits:** 1.5 (3 hours of lab work per week)

9. **Contact Hours:** 3 hours of lab work per week

10. **Total Marks:** 150

11. **Rational of the Course:**
    This course will cover how the client-server model of internet programming works and how the relevant internet programming tasks are accomplished. Subjects covered include basic web page development and an introduction to dynamic web page development using client-side scripting, server-side scripting, and database connectivity. This course is crucial to design and develop interactive web applications. Moreover, it will help to build tools that assist in automating data transfer over the internet. This is a required course for all the students enrolling B. Sc. Engg. in CSE program. The catalogue description of the course is,

    Course Content: Sessional based on the following topics:

    Introduction: What is a website, Types of websites, Technology used in website development, Different between web design and development, Domain and Hosting.

    HTML: HTML Basics, Tags, use of the tag, Properties and Element, Implementation of HTML tag, properties and element., Setting Background and Text Colors, Adding Local and Remote Links, Adding Internal Links with the Named Anchor Tag, Linking and Embedding Graphics, Creating Lists and Nested Lists, Creating and Modifying Tables.

XML and XHTML: XML basics; XHTML: Document Structure, Basic Tags, Text Formatting, Character Entities, Lists, Links, Images, Color, Tables, Forms, Image Maps, Meta tags, Framesets.

CSS: CSS Introduction, CSS Syntax, Classification, Font, Text Color & background, Padding, Border, Margin, Positioning.

JavaScript: Introduction, Control Structures, Functions, Arrays, and Objects, Web APIs with JavaScript (example: Google Ajax API), jQuery Basics, Using selectors and events.

Server-Side Technology: LAMP, Web application frameworks (example: Silverlight, Adobe Flex, Laravel), Web 2.0 and Web APIs, MVC framework.

PHP: Basics of About PHP, PHP Installation, PHP Syntax & PHP Variable, PHP Variable, PHP show/print output, PHP Operators, if (...else) statement, Switch statements, for statement, while statement, Do While statement, Arrays, File, File Upload, Cookies, Session, PHP Date, PHP Socketing.

Database: Database Basics, Types of Databases, Usage of Database, Database Schema.

MySQL Database: MySQL Database Basics, Features of MySQL Database, Database create and use, Table data type, create table and view, Insert, Update, delete, Logical Query, Constraint, Joining, Aggregate Functions, Have and group by clause, Complex Query of PHP and MySQL.

PHP and MySQL: Host Connection, Database Connection, inserting into table, Update Data from Table, View Data from Table, Delete Data from Table, Joining Data, Report Generation, Report Print, Session Authorization, log in and Log out, Hashing Password.

E-commerce: Introduction to E-business, Future of E-commerce

12. Course Objectives:

(a) Familiarization with various web development concepts such as HTML, XML, CSS, JavaScript, MySQL Database, etc.
(b) To learn and follow the analysis and design procedure of websites/android apps.
(c) To implement a project, suitable for and meets the requirements of the end-users.

13. Course Learning Outcomes (CLOs) and Mapping of CLOs with Program Learning Outcomes (PLOs)

a) CLOs

No.	Course Learning Outcomes (CLOs)	Bloom's Level (Optional)
CLO1	Recognize various web development concepts	C2, C4
CLO2	Reproduce the analysis and design procedure of websites/android apps	C1, P3
CLO3	Develop projects, suitable for and meets the requirements of the end-users	C5

b) Mapping of CLOs with PLOs

No.	CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
1	CLO1	X											
2	CLO2			X									
3	CLO3					X							

Part B

14. Course plan specifying content, CLOs, co-curricular activities (if any), teaching-learning, and assessment strategy mapped with CLOs

Course Plan

`	Topic	Teaching-Learning Methodology	Assessment Method	Corresponding CLOs
Week-01	Course overview, Introduction to HTML, XML, CSS, JavaScript, MySQL Database	• Lecture on the theoretical background and design principle		CLO-1
Week -02	Project topic selection	• Hands-on demonstration on the implementation • Group Discussion	Lab Performance Report	CLO-3
Week -03	Design and analysis of the projects	• Hands-on demonstration on the implementation • Group Discussion	Lab Performance Report	CLO-3

Week -04	Evaluating initial feasibility of the project	• Hands-on demonstration on the implementation • Group Discussion	Lab Performance Report	CLO-2
Week -05	Monitoring project progress (front-end)	• Hands-on demonstration on the implementation • Group Discussion	Lab Performance Report	CLO-2
Week -06	Monitoring project progress (back-end)	• Hands-on demonstration on the implementation • Group Discussion	Lab Performance Report	CLO-2
Week -07	Monitoring project progress (front-end)	• Hands-on demonstration on the implementation • Group Discussion	Lab Performance Report	CLO-2
Week -08	Monitoring project progress (back-end)	• Hands-on demonstration on the implementation • Group Discussion	Lab Performance Report	CLO-2
Week -09	Monitoring project progress (frontend-backend integration)	• Hands-on demonstration on the implementation • Group Discussion	Lab Performance Report	CLO-2
Week -10	Improving User Experience (UX/UI)	• Hands-on demonstration on the implementation • Group Discussion	Lab Performance Report	CLO-2
Week -11	Final Project Submission	• Group Discussion	Final Performance	CLO-3
Week -12	Presentation and evaluation of the project		Report Presentation	
Week -13	Viva-voce			

Part C

15. Assessment and Evaluation

l) Assessment Strategy

Class participation and attendance	10%
Class tests / Class assessment	20%
Term Final Examination (3 hours duration)	70%
Total	100%

2) Marks distribution:

a) Continuous Assessment: 70%
b) Summative: 30%

3) Make-up Procedures:

- Course teacher may arrange for makeup lab schedules if necessary.

Part D

16. Learning Materials

- **Recommended Readings:**

1. HTML, XHTML AND CSS
   Author: Shelly/Woods/Dorin
   Publisher: Cengage Learning

2. HTML AND CSS: DESIGN AND BUILD WEBSITES
   Author: Jon Duckett
   Publisher: Wiley

3. JAVASCRIPT & JQUERY INTERACTIVE FRONT-END WEB DEVELOPMENT
   Author: Jon Duckett
   Publisher: Wiley

- **Others**

1. Various online resources